

XINRONG ZONG

xinrong.zong@mail.polimi.it
+39 3242613083 | Milan, Italy, 20134

Education

Politecnico di Milano , Italy	Sep. 2024–Present
Master's studies in Chemical(Food) Engineering; Expected: 101/110	
Middle East Technical University , Turkey	Feb.–Aug. 2026
Erasmus+ Exchange Student, Department of Chemical Engineering; GPA: 3.60/4.00	
Hong Kong University of Science and Technology , Hong Kong	Feb.–Aug. 2024
Visiting internship, Department of Chemistry, Hong Kong; Grade: 92.5/100	
Northwest A&F University , China	Sep. 2020–Jun. 2024
Undergraduate studies in Food Science and Engineering; GPA: 85.7/100	

Research Experience

Molecular Design of Dynamic Bispidine-Based Coordination Polymers for Selective VOC Adsorption

Master's Thesis Researcher | Supervisor: Prof. Massimo Cametti

- **Investigating** ester-substituent engineering of chiral bispidine–Hg(II) coordination polymers to determine how local steric and chemical changes regulate crystal packing, framework dynamics, and VOC selectivity.
- **Designing** ligand variants through modification of the –COOR alkoxy substituents and evaluating their effects on interchain interactions, guest-accessible pockets, and molecular recognition toward MeOH, EtOH, and CHCl₃.
- Employing coordination-polymer synthesis, crystallization, desolvation, and competitive vapor adsorption, combined with PXRD, ¹H NMR, TGA, and SC-XRD, to establish structure–property relationships.

Deep Learning–Enhanced Electrochemical Sensor for Methyl Parathion Detection

Research Assistant | National Research Project No. 31901773

- Developed a **low-cost** electrochemical sensing framework for methyl parathion detection by integrating electrochemically activated screen-printed carbon electrodes (E-SPCEs) with machine-learning-assisted electrochemical impedance spectroscopy (EIS) analysis.
- Optimized and evaluated electrochemical **activation conditions** and analyzed electrode behavior using CV, DPV and EIS, together with XPS and Raman characterization to investigate changes in surface oxygen-containing functional groups, structural defects and electron-transfer kinetics.
- Developed a **hybrid CNN–SVM** classification pipeline for equivalent circuit model (ECM) recognition from EIS data, combining CNN-extracted image representations with 30 statistical, frequency-domain and geometric curve features, supported by resampling, cubic-spline interpolation and grid-search hyperparameter optimization.
- Achieved **88.4%** ECM classification accuracy on 502 EIS samples, substantially outperforming conventional SVM, Random Forest, standalone CNN and other benchmark models; electrochemical activation additionally produced a **21×** increase in MP response, reduced charge-transfer resistance by approximately **87.7%**, and achieved a 0.05 μ M detection limit.

Optimization of Electrochemical Aptasensor Stability for Pesticide Detection

Research Assistant | National Research Project No. 31901773

- Investigated key factors governing the stability of electrochemical aptasensors for acetamiprid detection by systematically evaluating electrode electro-activation, aptamer immobilization, and 6-mercaptop-1-hexanol (MCH) blocking conditions.
- Characterized electrode surface and interfacial properties using electrochemical impedance spectroscopy (EIS), cyclic voltammetry (CV), scanning electron microscopy (SEM), and X-ray photoelectron spectroscopy (XPS), linking changes in surface morphology, effective electrochemical area, Au–S binding sites, and aptamer adsorption to sensor performance.
- Optimized aptamer immobilization parameters by evaluating temperature and aptamer concentration, identifying excessive aptamer loading and enhanced physisorption as major sources of signal variability; aptamer concentrations around and above 1 μ M increased RSD from <5%→>10%.
- Improved sensor stability through MCH blocking optimization, reducing Rct variability from >15% to <7% while decreasing Rct from approximately 9.35 k Ω to 4.41 k Ω , demonstrating that prolonged MCH treatment effectively removes physically adsorbed aptamers.

Electrochemical Response Mechanisms of Phenolic Compounds on Electro-Activated Glassy Carbon Electrodes

Research Assistant

- Performed routine electrochemical and surface characterization of bare and electro-activated glassy carbon electrodes using CV, EIS, XPS and AFM; quantified changes in charge-transfer behavior, surface roughness and oxygen-containing functional groups.
- Analyzed and validated experimental data linking phenolic-compound structure—including hydroxyl-group number, aromatic-ring number and substituent properties—to adsorption behavior and electrochemical response.
- Completed supplementary validation experiments for catechol sensing, supporting a 46.23 nM LOD, 3.29% reproducibility RSD and 84.0–96.5% recovery in honey samples.
- Contributed to manuscript writing and publication, including data organization, figure preparation, journal submission, reviewer-response revision and additional experiments.

Publications

Published articles

Qingshuang Wei, **Xinrong Zong**, Yitao Lv, Chaoqi Wang, Jiacheng Wang, Min Zhang. Enhancing or not: What dominates the response signal of phenolic compounds on electro-activated glassy carbon electrode? *Journal of Electroanalytical Chemistry*, 967 (2024), 118457.

Yunying Yang, Sian Chen, Changqiu Zhang, Yanqing Li, **Xinrong Zong**, Yitao Lv, and Min Zhang. Subtle adjustment of the cyclic potential on electro-activated glassy carbon electrodes for sensitive sensing of methyl parathion. *Analytical Methods*, 16 (2024), 2522–2532.

Yanqing Li, **Xinrong Zong**, Sian Chen, Min Zhang. Research Progress on electrochemical paper-based analytical devices for the Detection of Pesticide Residues. *Food Science*, 2023, 15, 252–262.

Submitted manuscripts

Dual-Innovation in Methyl Parathion Detection: A Novel Electrochemical Activation Screen-Printed Sensor Coupled with Deep Learning Enhanced Equivalent Circuit Modeling. *Submitted; first author.*

Ignored Factors Influencing the Stability of Conventional Electrochemical Aptasensors. *Submitted; co-first author.*

Technical Skills

Electrochemistry: Electrochemical workstation; cyclic voltammetry (CV), linear sweep voltammetry (LSV), differential pulse voltammetry (DPV), Electrochemical Impedance Spectroscopy(EIS).

Materials characterization: NMR, UV-Vis, HPLC, XRD, XPS, Raman spectroscopy, SEM, TEM.

Programming and Software: Python, MATLAB, Origin, GraphPad Prism, Chemdraw, SPSS, Microsoft Office, Zotero, Photoshop.

Other software: Microsoft Office, Zotero, Photoshop.

Languages: Chinese (native); Italian(B2);English (IELTS 7.0).

Additional Experience

Food Inspector, Xi'an New District Market Supervision Administration Jun. 2023–Sep. 2023

- Conducted on-site food sampling and inspection in restaurants near primary and secondary schools.
- Performed sample analysis using gas chromatography, liquid chromatography, and microbial detection methods, and prepared reports in accordance with national standards (GB).

Leadership & Service

Project Leader, national-level College Student Innovation Project Jun. 2022

Head of the Learning Department, Student Union Jun. 2021–Sep. 2022

Outstanding Volunteer, Yangling Agri Sci & Tech Zone Marathon Apr. 2022

Scholarships & Funding

DSU Regional Scholarship, Lombardy, Italy (EUR 8,000/year; EUR 16,000 total) 2025–2026

Erasmus+ Mobility Grant, Politecnico di Milano (EUR 4,000) 2026

Third-class Professional Scholarship, Northwest A&F University (CNY 1,000) Jun. 2021